Shifted count comparison: analytic replacement and domain audit

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Prepared with AI assistance. This note records conditional analytic reductions and bounded checks; it does not assert a complete Berry–Esseen proof.

1 A count comparison adapted to the available strip

The integer used to compare an iid sum with a binomial sum is a choice, not a property of the law. Choosing it to fit the conditional support removes one reason for requiring a half-amplitude strip. This section proves that change and describes all other estimates that must change with it. It does not assert a complete numerical comparison on the new domain.

Let $Y = B+U$, where B is Bernoulli with parameter $0 < p < 1$, $q = 1-p$, and $\mathbb{E}(U \mid B) = 0$. Denote the conditional laws by U_0, U_1, and suppose

$$|U_j| \le A, \qquad U_0 \le a_0, \qquad U_1 \ge -a_1, \qquad a_0, a_1 \ge 0, \quad a_0 + a_1 \le d_0 \le A.$$

Write $v = pq$, $s = q\mathbb{E}U_0^2 + p\mathbb{E}U_1^2$, $\lambda = ns$, $\ell = (v+s)^{-1/2}$, $\rho = \mathbb{E}|\ell(Y-p)|^3$ and $\rho_B = (1-2v)/\sqrt{v}$. The actual raw threshold is H and $x = np + H/\ell$. Set $\zeta = 1 - d_0$.

1.1 Choice of phase and the exact identity

Assume $A + 2d_0 < 1$ and define

$$l = \frac{A - 1 - 2d_0}{2}, \qquad h = l + 1, \qquad \epsilon = \frac{1 - A - 2d_0}{2} > 0. \tag{1}$$

Every $x \in \mathbb{R}$ has a representation $x = k+u$ with integer k and $u \in [l, h]$, for example $k = \lfloor x - l \rfloor$; either choice is allowed at a shared endpoint. This assertion alone does not ensure that k is a legal or interior binomial count. For example, if $p \in [p_-, p_+]$, $|H/\ell| \le X_*$ and $n \ge N$, the sufficient conditions

$$Np_- - X_* - h > 0, \qquad N(1 - p_+) - X_* + l > 0$$

ensure $1 \le k \le n - 1$.

Notice that $d_0 \le A$ and $A + 2d_0 < 1$ imply $d_0 < 1/3$, $A < 1$ and $-\zeta < l < h < \zeta$: indeed $h < \zeta$ is equivalent to $A < 1$, while $l > -\zeta$ follows from $A + 1 - 4d_0 \ge 1 - 3d_0 > 0$.

For a legal count define $\delta = k - np$, $\pi_j = \binom{n}{j} p^j q^{n-j}$, with zero extension, and F_B its CDF. Let V_j be a sum of j independent copies of U_1 and $n - j$ of U_0. Conditional independence gives exactly

$$Q := \mathbb{P}\Big\{ \sum Y_i \le k + u \Big\} - F_B(k) = -\sum_{i=0}^{k} \pi_{k-i} \mathbb{P}\{V_{k-i} > u + i\} + \sum_{i=1}^{n-k} \pi_{k+i} \mathbb{P}\{V_{k+i} \le u - i\}.$$

The outgoing event is strict and the incoming event closed. Put $c = C_E$, $a = \sqrt{n}\{\mathbb{P}(\sum Y_i \le x) - \Phi(H/\sqrt{n})\} - c\rho$, and

$$C_B = \frac{\sqrt{n}}{\rho_B}\{F_B(k) - \Phi(\delta/\sqrt{nv})\}, \qquad f(u) = \frac{\Phi((\delta+u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k}.$$

Direct subtraction, with no phase-size assumption, gives

$$a + \rho_B(c - C_B) = \sqrt{n}\pi_k\{Q/\pi_k - f(u)\} + \mathcal{R}, \tag{2}$$

where

$$\mathcal{R} = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + c(\rho_B - \rho).$$

1.2 Crossings and curvature on the longer phase interval

Couple all $2n$ conditional variables independently and set $V_j = \sum_{i\le j} U_{1,i} + \sum_{i>j} U_{0,i}$, extending constantly outside $[0, n]$. Then $j + V_j$ has increments at least ζ inside the legal range and one outside. Consequently

$$K = \max\{j \in \mathbb{Z} : j + V_j \le k + u\}, \quad R = K - k, \qquad \mathbb{P}\Big\{\sum Y_i \le k + u\Big\} = \mathbb{E}F_B(k + R).$$

For real z, define the finite sum $T(z) = \sum_{j\ge1}(z-j)_+$. Let $\chi = (n+1)p - k$ and $m_B = \max_j \pi_j$. For every integer r,

$$F_B(k+r) - F_B(k) - r\pi_k \le \frac{m_B}{(n+1)v}\left\{T(\chi)r_+ + \chi_+\binom{r_-}{2} + \binom{r_-}{3}\right\}. \tag{3}$$

Indeed, the binomial difference identity bounds the positive part of $\pi_j - \pi_{j-1}$ by $m_B((n+1)p - j)_+/((n+1)v)$, including its zero extension. For $r > 0$, double summation gives weights $r - i + 1 \le r$ on the differences at $k + i$; summing their positive bounds gives $rT(\chi)$. For $r = -m < 0$, the weights are $m - 1 - i$, $0 \le i \le m - 2$, on differences at $k - i$. The bound $(\chi + i)_+ \le \chi_+ + i$ gives the other two terms. There is no restriction $\chi < 2$ in this proof.

More generally suppose $-\zeta < l \le u \le h < \zeta$, $0 < \zeta \le 1$, and write $t = \mathbb{E}V_k^2$. Crossing monotonicity gives

$$\mathbb{E}R_+ \le \frac{t}{(\zeta-u)^2}, \qquad \mathbb{E}\binom{R_-}{2} \le \frac{t}{[\zeta + \min(u, \zeta/3)]^2}, \qquad \mathbb{E}\binom{R_-}{3} \le C_3 t\sqrt{A^2 + 3t}, \tag{4}$$

where

$$C_3 = \sup_{j\in\mathbb{Z},\, j\ge2} \frac{j^3 - j}{6(\zeta j + l)^3}.$$

For the first estimate, $R = r \ge 1$ implies $-V_k \ge \zeta r - u > 0$. The difference $(\zeta r - u)^2 - r(\zeta - u)^2$ equals $(r-1)(\zeta^2 r - u^2) \ge 0$. For $R = -m \le -2$, the first unaccepted count gives $V_k > \zeta(m-1) + u > 0$. For $u \le \zeta/3$, $j(j+1)/(2(\zeta j + u)^2)$ decreases on $j \ge 1$, since its derivative has the sign of $j(2u - \zeta) + u \le 0$. For larger u, lower it to $\zeta/3$. This proves the second estimate. The same strict crossing with $j = m - 1 \ge 2$ proves $\binom{R_-}{3} \le C_3(V_k)_+^3$. Finally, independence and centering give $\mathbb{E}V_k^4 \le A^2 t + 3t^2$, and Cauchy–Schwarz proves the last bound.

For $A = 3/5$, $d_0 = 9/125$, the interval is

$$[l, h] = [-34/125, 91/125], \qquad \zeta = 116/125, \qquad C_3 = \frac{1953125}{7739786}. \tag{5}$$

To verify the last value, the derivative of $(j^3 - j)/(6(\zeta j + l)^3)$ has the sign of $3lj^2 + 2\zeta j - l$. For these constants it is positive at 2, negative at 3, and has just one positive zero. Thus only the integers $2, 3$ can maximize it; the value at 3 is the larger. For other $l < 0$, the same argument reduces the integer maximum to 2 and the adjacent integers to $(\zeta + \sqrt{\zeta^2 + 3l^2})/(-3l)$. If $l \ge 0$, its supremum is $1/(6\zeta^3)$.

The phase supremum still has three candidates. For $0 < \zeta \le 1$ and $[l, h] \subset (-\zeta, \zeta)$, put

$$F(u) = \frac{T(b+u)}{(\zeta-u)^2} + \frac{(b+u)_+}{[\zeta + \min(u, \zeta/3)]^2}.$$

Its maximum is the larger endpoint value, or the value $1/[4(\zeta-b)]$ when

$$\zeta > b, \quad l \le \zeta - 2b \le h, \quad \zeta - 2b \le \zeta/3, \quad \zeta - 2b \le 1 - b.$$

Here is the point at which the old $|u| \le 1/2$ proof needs care. Each active summand $(b+u-j)/(\zeta-u)^2$ has strictly positive second derivative: its numerator is twice $3(b+u-j)+2(\zeta-u) > 0$. Below $\zeta/3$, the second derivative of the other term has sign $3b+u-2\zeta$. Whenever $T(b+u) > 0$, this is greater than $3 - 2u - 2\zeta > 3 - 8\zeta/3 \ge 1/3$. Thus their sum is convex there even on the longer phase interval. Above $\zeta/3$ the second term is piecewise linear. All derivative jumps at $b+u = 0, 1, 2, \ldots$ and at $u = \zeta/3$ are nonnegative. An interior maximum can therefore occur only where $T = 0$ and the derivative $(\zeta - 2b - u)/(\zeta+u)^3$ vanishes. The displayed eligibility and value follow. Continuity includes coincident breakpoints and constant pieces.

1.3 The marked crossing needs no half-amplitude bound

Let $D_i = U_{1,i} - U_{0,i}$ and $\vartheta = \mathbb{E}(D_i)_+ = \mathbb{E}(D_i)_-$. For $0 < \eta \le 1$, write $R_f = \mathbb{E}\{u + \eta - V_k\}$, where braces denote fractional part. The deleted sums have laws $A_0 = kU_1 + (n-k-1)U_0$ and $A_1 = (k-1)U_1 + (n-k)U_0$. Then

$$\mathbb{E}R \le u + \eta - R_f + \frac{\vartheta}{\eta} \sum_{i \ge 1} \{\mathbb{P}(A_0 \le u + d_0 - \zeta i) + \mathbb{P}(A_1 > u + \eta + i - A)\}. \tag{6}$$

To prove it, set $z = u - V_k$ and use the marks

$$Z_f = \sum_{i=1}^{n-k} (D_{k+i})_- \mathbf{1}_{\{V_k \le u - \zeta i\}}, \quad Z_b = \sum_{i=1}^{k} (D_{k-i+1})_+ \mathbf{1}_{\{V_k > u + \eta + i\}}.$$

On $R > z + \eta$, if $R > 0$ the crossing implies $z \ge \zeta R$, so all forward marks through R are active and $R \le z + Z_f$. If $R < 0$, $z < R - \eta$ activates all backward marks through $-R$, and the accepted count gives $R \le z + Z_b$. The case $R = 0$ is impossible on this event. Thus if $h = R - \lfloor z + \eta \rfloor > 0$, then $Z_f + Z_b > h - 1 + \eta \ge \eta h$. This yields the pathwise inequality $R \le \lfloor z + \eta \rfloor + (Z_f + Z_b)/\eta$. Deleting a forward pair uses $U_0 > U_1 \ge -d_0$ on its active mark; deleting a backward pair uses $U_1 \le A$. Independence from the deleted pair then gives (6). The bound $|R| \le 1 + (|V_k| + |u|)/\zeta$ justifies expectations. No phase restriction has been used.

On the shifted interval, the first individual distances are

$$1 - 2d_0 - h = 1 - A + l = \epsilon > 0.$$

For the optional common-deleted-sum estimate they become $1 - 3d_0 - h = 1 - A - d_0 + l = \epsilon - d_0$; hence that extra shortcut requires $A + 4d_0 < 1$. The concrete values in (5) give $\epsilon = 16/125$ and $\epsilon - d_0 = 7/125$. The fourth-moment tail denominators for marks $i \ge 3$ are $(5/2)\zeta - A - h = 124/125$ and $(5/2)\zeta - A + l = 181/125$, both positive. The conditional spans are at most $A + d_0 = 84/125 < 1$, so the harmonic fractional-part estimate applies. These are domain checks, not a numerical upper bound for the full error in (2).

1.4 Other changes that the shifted anchor requires

The count must be held fixed while differentiating f. A Stirling estimate $\sqrt{n}\pi_k = \phi(0)v^{-1/2}\exp\{-\delta^2/(2nv) + e\}$, $n|e| \le Z(D)$, is used on the enlarged count displacement

range. On $|\delta| \le D$, $|u| \le a$, the usual derivative bounds remain valid with this new a, without requiring $a \le 1/2$. In particular, the term $(2Da + a^2)/(2v_-) + Z(D)$ must be recomputed. The positive-phase signed Gaussian estimates also hold on any $[a, b] \subset [0, \infty)$: their proof uses convexity on the segment from 0 to the actual u, not $u \le 1/2$. The negative-phase estimate has the same property.

For tiny noise, the shifted-quartic argument can still use $r = zt$ and its original quadratic lower bounds. If the negative phase has maximum magnitude $a_- = -l > 0$, the far-end Cantelli gate must be

$$\frac{a_-^2}{t_* + a_-^2} - h_f a_- > c_- t_*, \qquad z_- t_* < a_-. \tag{7}$$

The other gates are unchanged: $J(1, -h_f, z_-) > c_-$, $z_-^2/(1 + z_-^2 t_*) - h_f z_- > c_-$, and $2z_-/(1 + z_-^2 t_*)^2 > h_f$. Indeed $r^2/(t + r^2) - h_f r$ has positive derivative at $r = z_- t$; its second derivative changes sign once, so any later stationary point is a maximum. Its minimum on $[z_- t, a_-]$ is at an endpoint. The left endpoint divided by t decreases with t, and the right endpoint is bounded by (7). On the positive side replace $z_+ t_* < 1/2$ by $z_+ t_* < h$. The quartic and the bound $f(u) \ge l_f u$ then give the same proof. These conclusions still require their numerical gates; changing the anchor does not itself control noise tending to zero.

The binomial ratio shortcut $p_+ + \Delta < 2$ need not hold on the enlarged displacement range. One safe replacement is $\pi_{k+i}/\pi_k \le M_B/m_0$ whenever $\sqrt{n}\, m_B \le M_B$ and $\sqrt{n}\pi_k \ge m_0 > 0$. All-probability Stirling constants use $|1 - 2r|$, not $1 - 2r$, on their count-frequency segment. Likewise set $r_0 = \min(p, q)$ in the variance bound $|s_1 - s_0| \le s/r_0$.

The absolute cubic needs no reflection. If $d_0 < \min(p, q)$, its two conditional signs are fixed and

$$\mathbb{E}|Y - p|^3 \ge v(1 - 2v) + (3r_0 - d_0)s.$$

To see this, expand the cubic and bound $p\mathbb{E}U_1^3 - q\mathbb{E}U_0^3 \ge -pa_1 s_1 - qa_0 s_0$; the coefficients of qs_0 and ps_1 are respectively $3p - a_0$ and $3q - a_1$, both at least $3r_0 - d_0$. Hence

$$(v + s)^{3/2}(\rho - \rho_B) \ge s\left\{3r_0 - \tfrac{3}{2}(1 - 2v) - d_0 - \frac{3(1 - 2v)s}{8v}\right\}.$$

The same signed-multiplier rule remains necessary if this bracket is negative.

Finally, if the mass upper bounds are not normalized to one, the whole-count inverse-index term is

$$I(\lambda) = \frac{M_B \omega_+}{v_-}\{\sup F + C_3\sqrt{A^2 + 3T_U}\} + M_k a B_1 + \frac{M_k a P_1}{\lambda} + \kappa,$$

where $\sqrt{n}\pi_k \le M_k$ and $\mathcal{R} \le \kappa\lambda/n$. Thus a replacement domain must recompute both mass bounds, the own-count variance factors, all Gaussian constants, the cubic repair and marked tails.

1.5 Spend the available Bernoulli gap

Schulz's dissertation, Theorem 1 (printed page 15), equations (4) and (5), gives the strict finite-index Bernoulli bound for all $0 < p < 1$, and for $1/3 \le p \le 2/3$ the sharper bound

$$C_B < c(p) := \frac{\phi(0)(3 + |1 - 2p|)}{6(p^2 + q^2)}.$$

Here the statement concerns every threshold, since its source bounds the Kolmogorov distance. Put $p_* = (4 - \sqrt{10})/2$. Elementary expansion gives

$$g(p) := \rho_B(C_E - c(p)) = \frac{2C_E(\min(p, q) - p_*)^2}{\sqrt{pq}} \ge 0. \tag{8}$$

Outside $[1/3, 2/3]$, use $g(p) = 0$ and the strict universal Bernoulli bound. Thus any witness with $a \geq 0$ has left side of (2) strictly greater than $g(p)$. It is sufficient to bound its right side by $g(p)$, with a non-strict inequality. At zero noise, a maximizing Bernoulli threshold is an atom; the strict finite-index theorem excludes it directly. No separate unproved strictness assumption is needed.

Suppose on a probability interval I and noise interval $[L, U]$, $L > 0$, the comparison has been bounded by

$$\sqrt{n}\pi_k \lambda B(\lambda) + \lambda I_0(\lambda)/n, \quad m_0 \leq \sqrt{n}\pi_k \leq M_k, \quad I_0(\lambda) = A_0 + A_1/\lambda, \quad A_1 \geq 0,$$

where B is convex and all constants are fixed on the cell. Define

$$\Gamma(\lambda) = \max\{m_0 B(\lambda), M_k B(\lambda)\} + \frac{\max(0, I_0(\lambda))}{N}.$$

It is convex, since positive multiples and finite maxima preserve convexity. For any $0 \leq g_I \leq \inf_{p \in I} g(p)$, the two endpoint inequalities

$$\Gamma(L) \leq g_I/U, \qquad \Gamma(U) \leq g_I/U \tag{9}$$

give a contradiction: the comparison error is at most $\lambda\Gamma(\lambda) \leq \lambda g_I/U \leq g_I \leq g(p)$. This permits either sign of B and I_0. Subtracting g_I/λ from the old score would not preserve the same convexity argument.

The conclusions are analytic replacements for the count interface. They allow the 0.6 strip to replace half-amplitude in its domain assumptions, but only a complete numerical comparison (including the small-noise part) could remove half-amplitude confinement from the final proof dependency graph.